Rigorous Evaluation of LLM Components in HPC Research

Daniel Nichols*, Abhinav Bhatele, Harshitha Menon***

*Lawrence Livermore National Laboratory **University of Maryland

This work is performed under the auspices of the U.S. Department of Energy by Lawrence Livermore National Laboratory under Contract DE-AC52-07NA27344.

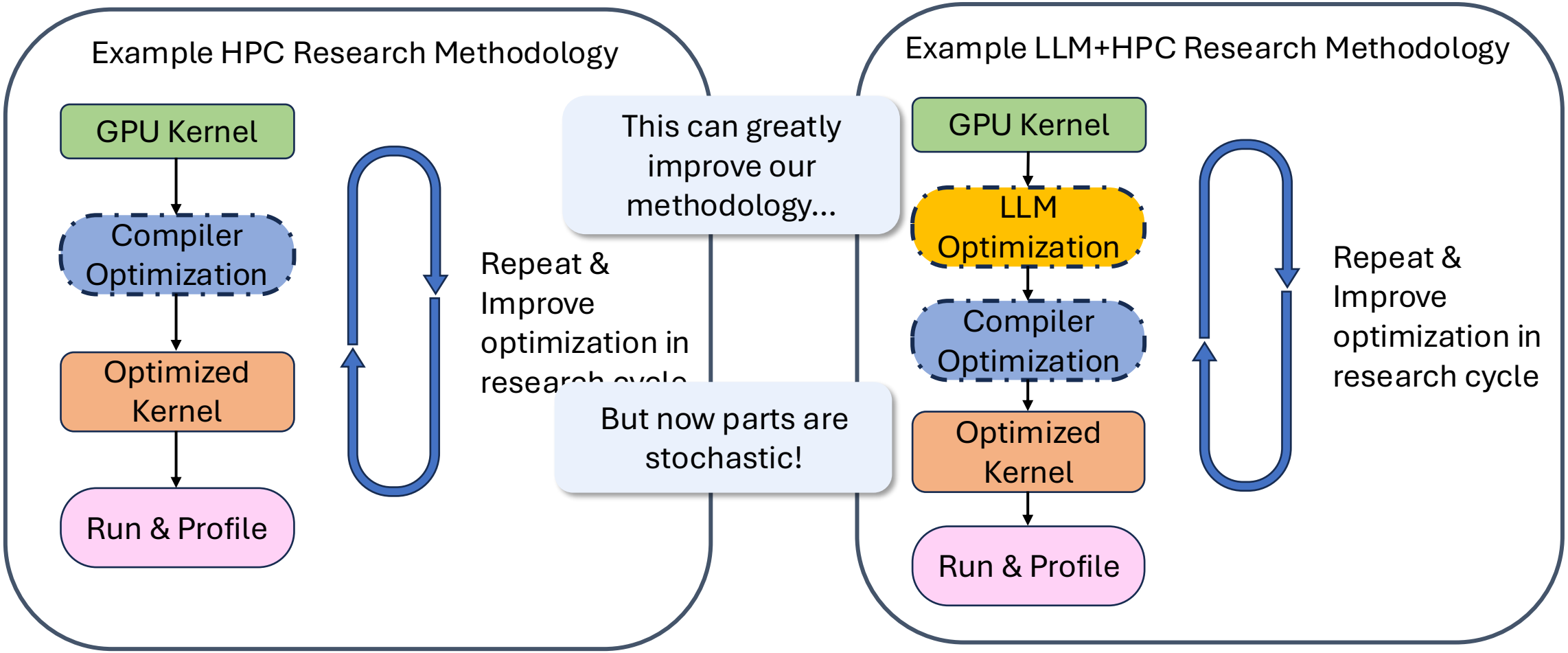
Section 1: Introduction

- Tutorial Objectives
  - Learn how to properly evaluate LLMs in HPC and scientific research
  - Learn the dangers and common pitfalls from using LLMs in research
- Overview
  - Intro to LLMs
  - How to Lie with LLM Results
  - Experimental Design for Stochastic Components
  - Transparency and Reproducibility
  - Human Validation: When and How
  - Baselines, Metrics, and Ablations
  - HPC and Scientific Computing Specifics

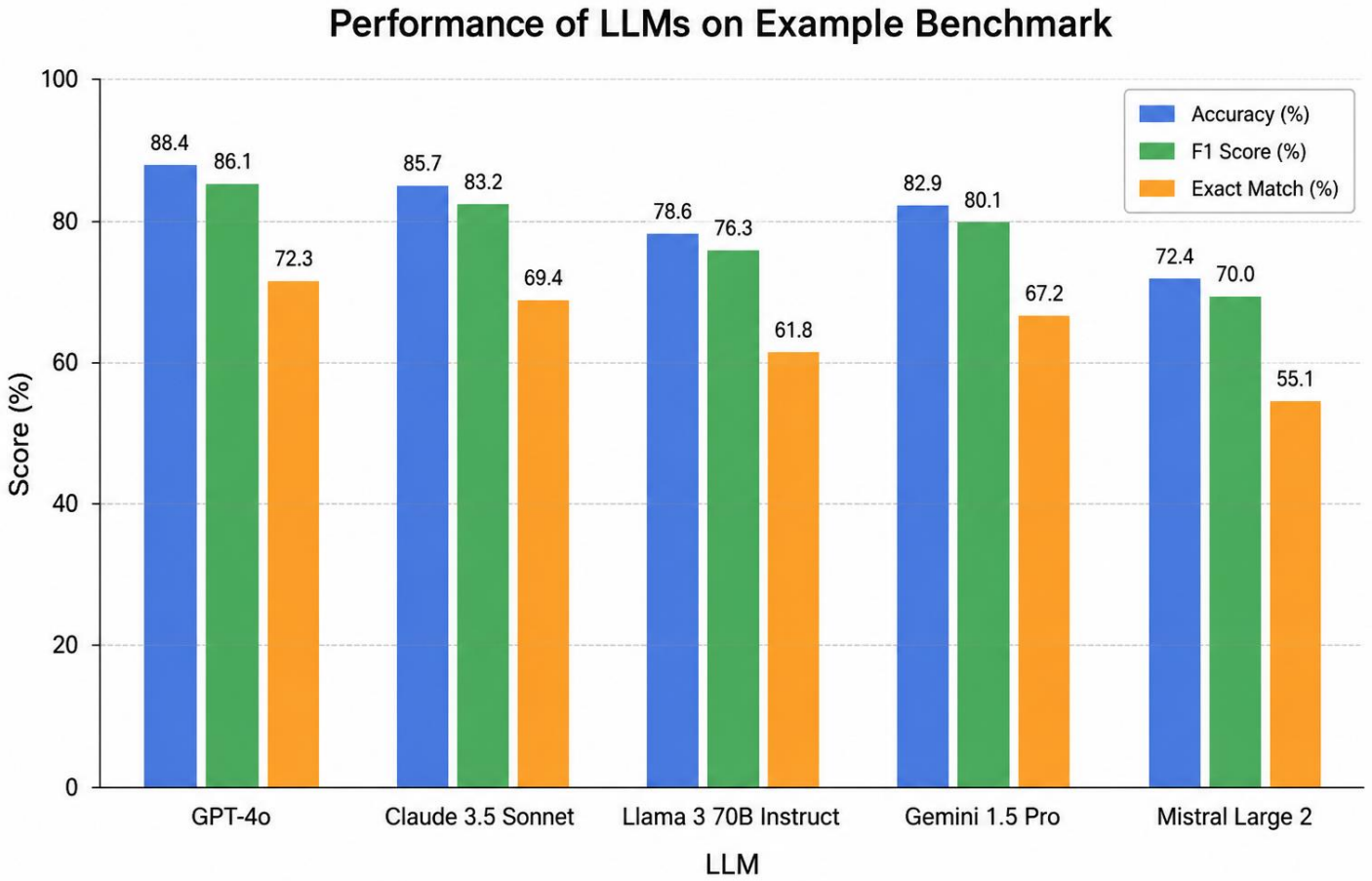
Tutorial site

https://parallelcodefoundry.github.io/tutorials/isc-2026-llm-eval/

Why this tutorial?



How should we evaluate LLMs and report results?



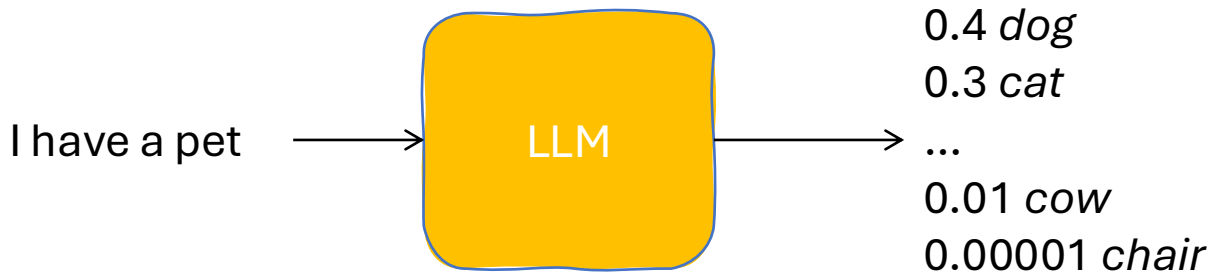
Section II: Introduction to LLMs

- Goal for this section: shared vocabulary for using LLMs
- You should be able to answer:
  - What are tokens?
  - What is text generation?
  - Where does randomness come from?
  - What is a model?

- Overview
  - What is an LLM?
  - What are tokens?
  - Text Generation
  - Sampling: How Randomness is Introduced
  - Randomness in LLMs

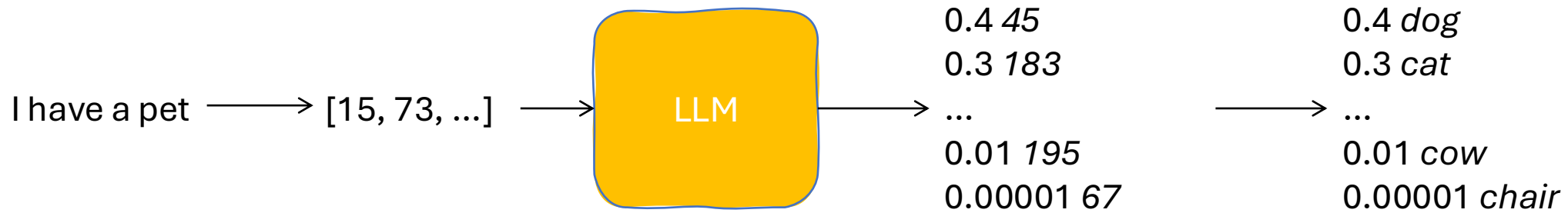
What is an LLM?

- Large Language Models are neural networks based on the transformer architecture that are state-of-the-art for modeling sequential data (e.g. text, language)
- Often trained for causal language modeling
  - Input: text sequence → output: probabilities of next token



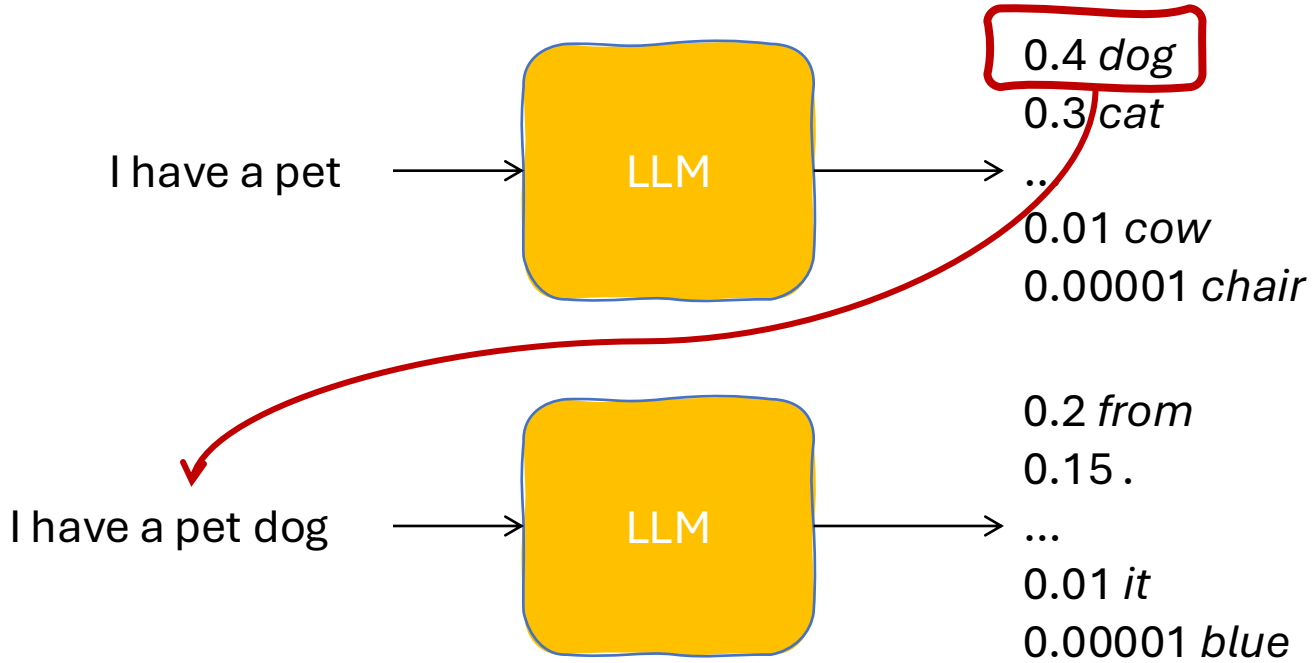
What are tokens?

- LLMs learn from integer sequences, not text
- *Tokenization* is the process of mapping text to integer sequences
- LLMs take sequences of tokens as input and outputs token probabilities
- Tokens *do not* correspond 1-1 with words or characters



Text Generation

- Auto-regressive text generation is the process of repeatedly sampling the next token to generate sequences



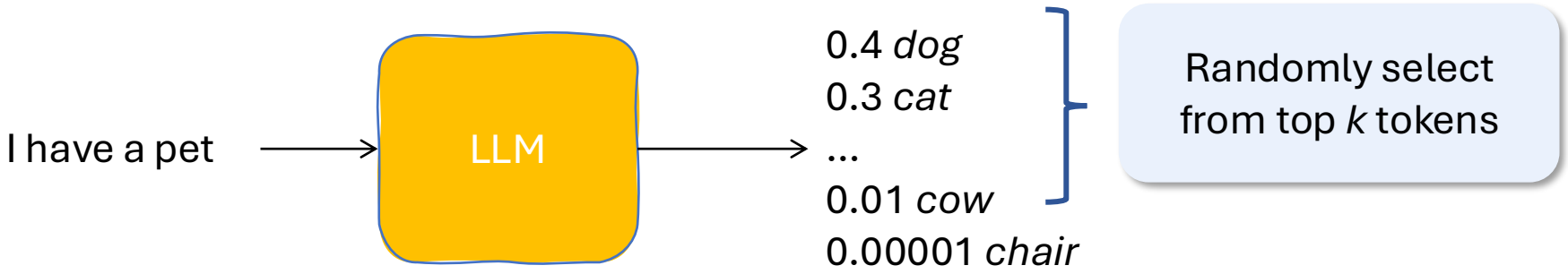
Text Generation

- Auto-regressive text generation is the process of repeatedly sampling the next token to generate sequences
- Stop when the model generates stop or EOS token
- Note: early differences in sequences can cascade into very different outputs

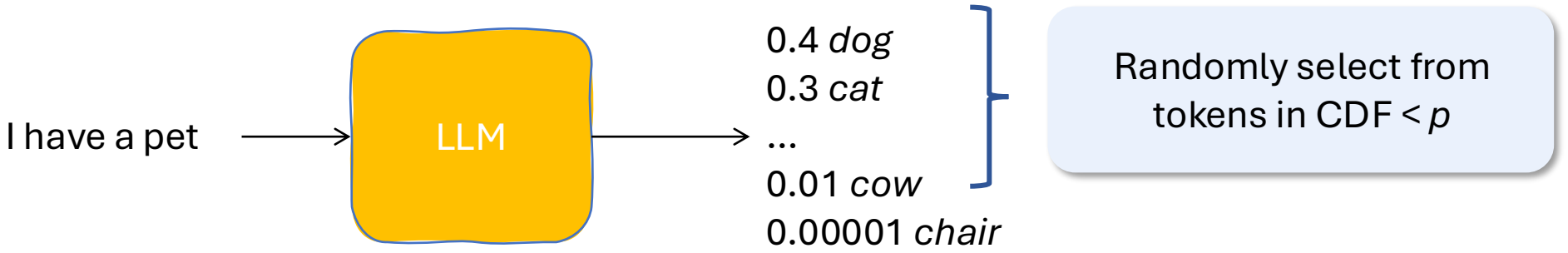
Sampling: The Introduction of Randomness

- Using the most probable next token leads to bad, repetitive generations (A. Holtzman et al. “The Curious Case of Neural Text Degeneration”)
- Randomly sampling yields much better results
- Popular sampling strategies
  - Top-K
  - Top-P or Nucleus Sampling
  - Beam search
- Temperature often used to scale probabilities before sampling

Top-K Sampling



Top-P Sampling



Randomness in LLMs

- Transformers with fixed weights and inputs are deterministic in the log-probs they output *, **
- Randomness is introduced by:
  - Prompt variance: multiple ways of prompting the same question, e.g. “Find the square root of N” vs “What multiplied by itself equals N?”
  - Sampling: Using randomness to sample the next token in text generation

* Parallelism, floating point arithmetic, and MoEs can cause non-determinism in some implementations

** Commercial/closed models may change system prompts and/or weights without notice

Demo 1: LLM Randomness

- See demo 1 to try different sampling strategies and their impact on generation outcomes

Section III: How to Lie with LLM Results

- Goal for this section: understand where LLMs can go wrong when used in research components

- Overview
  - Lie 1: Not considering randomness
  - Lie 2: Conflating formatting abilities with technical abilities
  - Lie 3: Cherry picking examples
  - Lie 4: Benchmark leakage / data contamination
  - Lie 5: Not considering other axes of evaluation
  - Lie 6: LLM-as-a-judge, rubric drift
  - Lie 7: Human-as-a-judge, inconsistent evaluation
  - Lie 8: Unreported human intervention
  - Lie 9: Not reporting cost / GPU hours

Lie #1: Not Considering Randomness

- LLM outputs are *random*; reporting metrics for a single output can be misleading
- Often only sampling variance is covered
  - Rarely we're benchmarking LLMs on prompts, but rather *problems*
  - Prompt variance matters! (Y. Zi et al. "More Than a Score: Probing the Impact of Prompt Specificity on LLM Code Generation")
- What should be done
  - Do multiple generations and report: *What is the probability/likelihood that a model outputs the correct solution?*
  - Many works/benchmarks employ this through probabilistic metrics e.g. mean accuracy, pass@k

Lie #2: Conflating Formatting Abilities with Technical Abilities

- If we are testing an LLM/Agent on its ability to do task *X*, then we don't want to conflate performance on *X* with performance on formatting outputs for task *X*.
- An example: You task an LLM with solving a math problem and outputting the final number; your script checks accuracy with `int(llm_answer) == ground_truth`.
  - The LLM would get this wrong if it generated "The answer is …"
  - Even if it knew the answer we've marked it as wrong
- Give LLMs the benefit of the doubt in parsing outputs, unless your task *X* includes formatting as part of the task

Lie #3: Cherry Picking Examples

- Another facet of random outputs from LLMs is that we must report all results
- Only reporting the “best” results can mislead to thinking the LLM is doing better than it actually is
- What should be done:
  - Sampling parameters need to be transparent: If the LLM tool only works for some sampling parameters, this is *ok*, but it needs to be reported

Lie #4: Benchmark Leakage and Data Contamination

- If your benchmark is in the training set, then the model could be over-fitting the answers
- You're testing how well the model can memorize your problems and not actually solve them
- This is only okay if your benchmark is *exhaustive*, which is almost never the case
- What should be done:
  - Even if your questions are private, make sure they are not too similar to existing public sources

Lie #5: Ignoring Important Axes of Evaluation

- To avoid previous lies, practitioners often sample many outputs and, thus, need easy-to-automatically-compute metrics for evaluation
- This often leads to Lie #3: using too simple of metrics that don't paint the whole picture
- Consider evaluating LLMs on generating MPI code

This is functionally correct, but not really correct

```
def compute_and_share(data):
  output = compute(data)
  for rank in other_ranks:
    mpi_send(output, rank)
```

This is what we want to see the LLM generate

```
def compute_and_share(data):
  output = compute(data)
  mpi_bcast(output)
```

Lie #6: LLM-as-a-Judge, Rubric Drift

- Rubric drift: small changes in prompts and judge models can drastically alter results
- LLM judging often conflates style and correctness
- LLM-based evaluation is a useful new research tool, but is *insufficient* and *potentially dangerous* for scientific evaluation

Lie #7: Human-as-a-Judge, Inconsistent Eval

- Humans are just as prone to evaluation biases as LLMs
- Evaluation criteria not public or precise. Evaluation data not public
- As with lie #6, human judgement is *insufficient* and *dangerous* in scientific results reporting
- What should be done:
  - Social science fields have best practices for using humans to evaluate / rank / label data. It is still very difficult to implement correctly.

Lie #8: Unreported Human Intervention

- LLM / Agent pipeline relied on human intervention somewhere and it is not reported
  - Data cleaning between components
  - Prompt design and/or variation
  - Sampling parameter and/or model choice changes
- What should be done:
  - For better reproducibility – study and document human interventions

Lie #9: Not Reporting Cost / GPU Hours

- API costs or GPU time when self-hosting are imperative to report, particularly when comparing with other approaches
- If an approach yields better results, but costs significantly more this is of interest to users of the approach
- What should be done:
  - All costs and GPU hours need to be reported
  - Costs fluctuate, so token counts should also be reported.

Section IV: Experimental Design for Stochastic Components

- Goal for this section: how to design experiments that rely on stochastic subcomponents

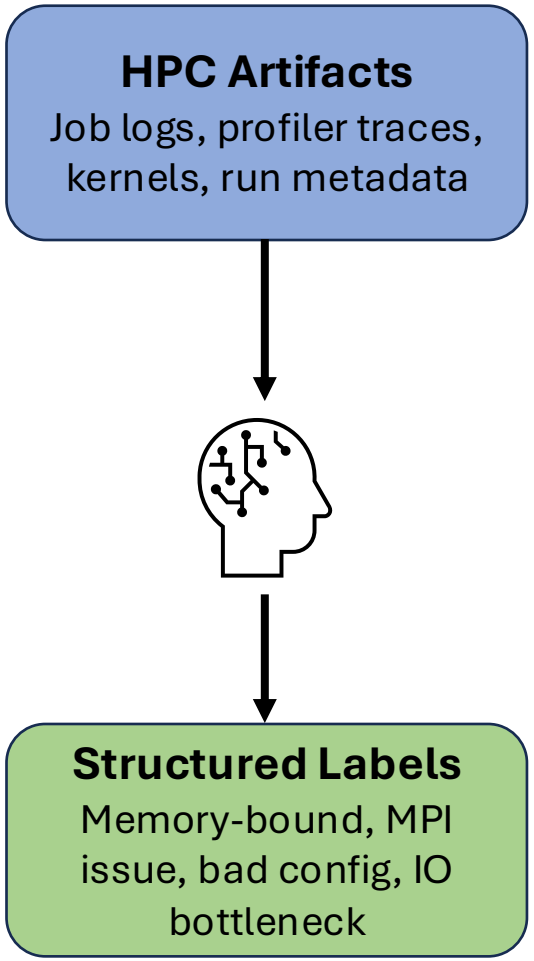
- Overview
  - How are LLMs used in experiments
  - Why careful consideration is important
  - Structured inputs/outputs; mitigating stochasticity
  - Testing individual subcomponents
  - Ablation studies on system
  - Human validation

How are LLMs used in experiments

- Annotation
- Judge
- Synthesis
- Generation
- Tools
- Benchmarking
- Subject

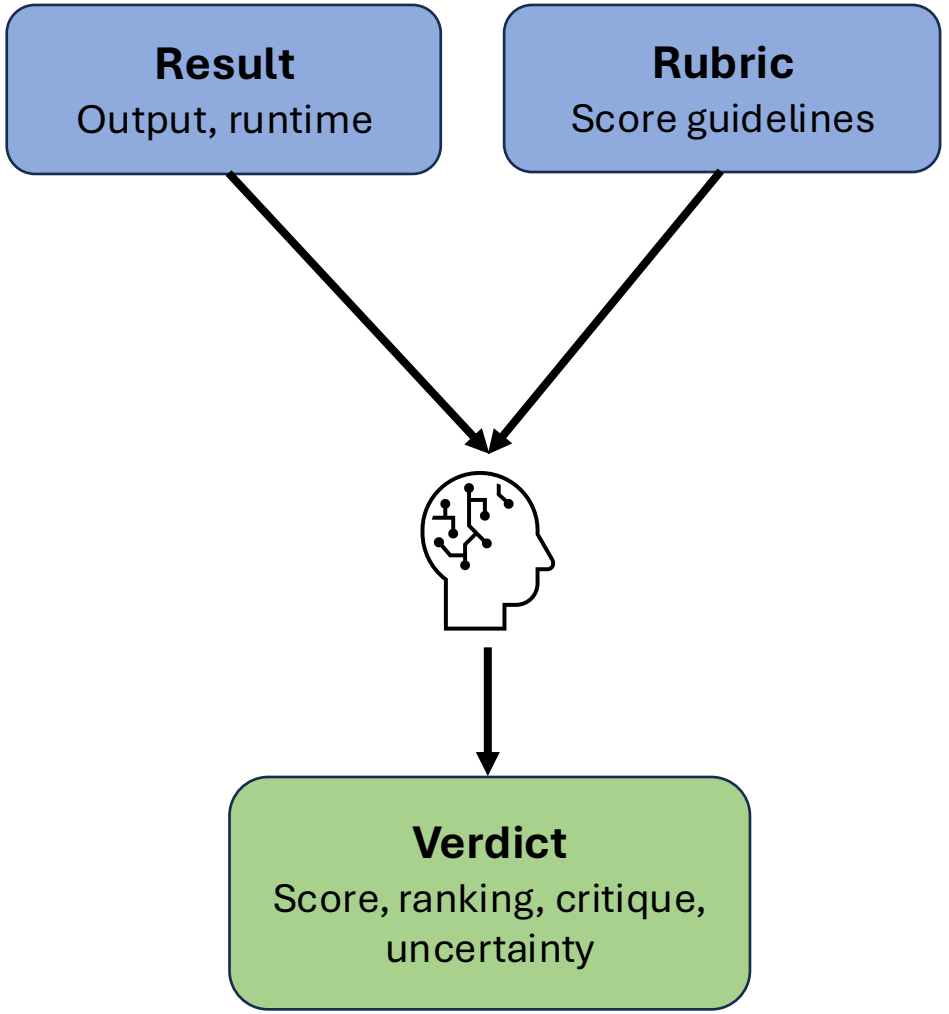
How are LLMs used in experiments: Annotation

- LLMs label research artifacts such as code, performance data, parallelism type, etc.
- Can apply a predefined coding guide or suggest candidate codes.



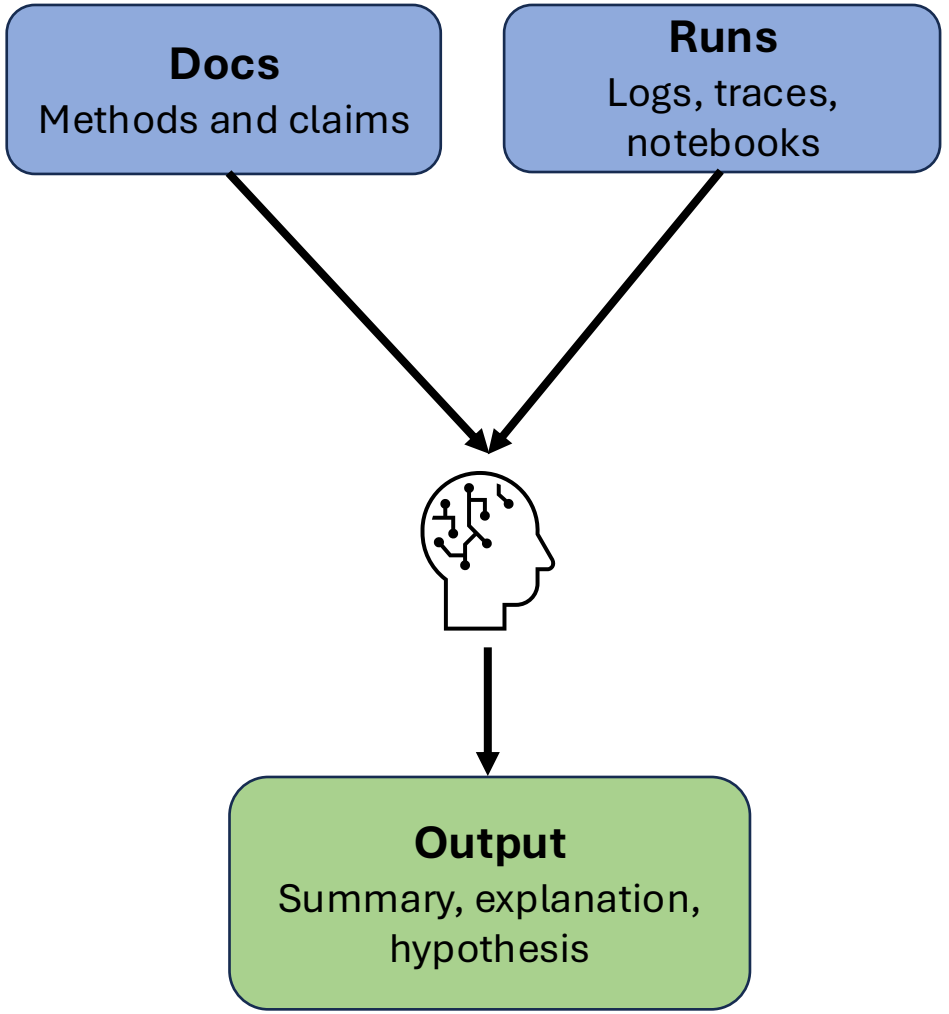
How are LLMs used in experiments: Judge

- LLMs score, rank, or compare HPC artifacts against a rubric, such as whether an optimization is plausible, a generated explanation is correct, or a run result supports a claim.



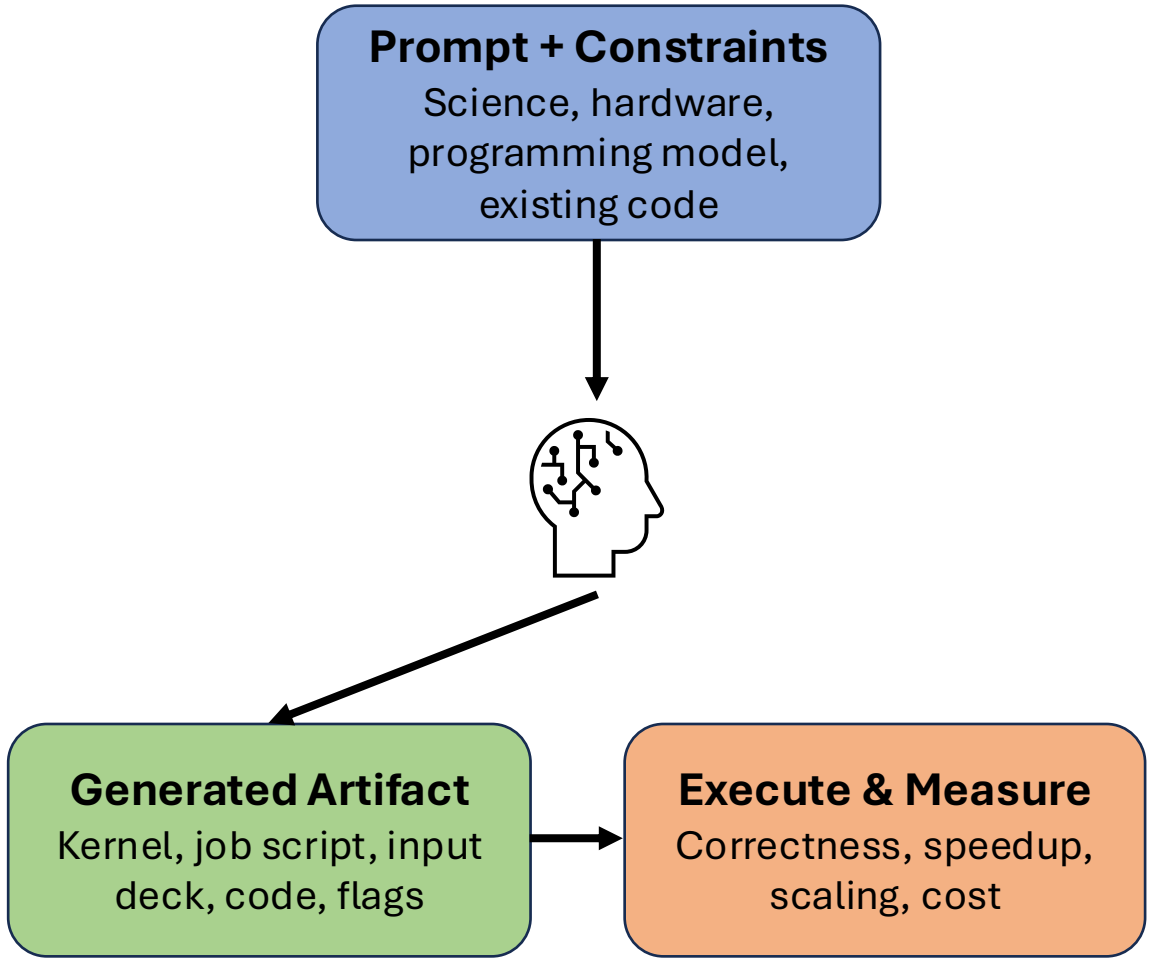
How are LLMs used in experiments: Data Synthesis

- LLMs combine evidence across papers, documentation, job outputs, profiler reports, and experiment notes to produce summaries or hypotheses.
- Generate synthetic kernels, input decks, configurations, or expected failure cases for later testing and evaluation.



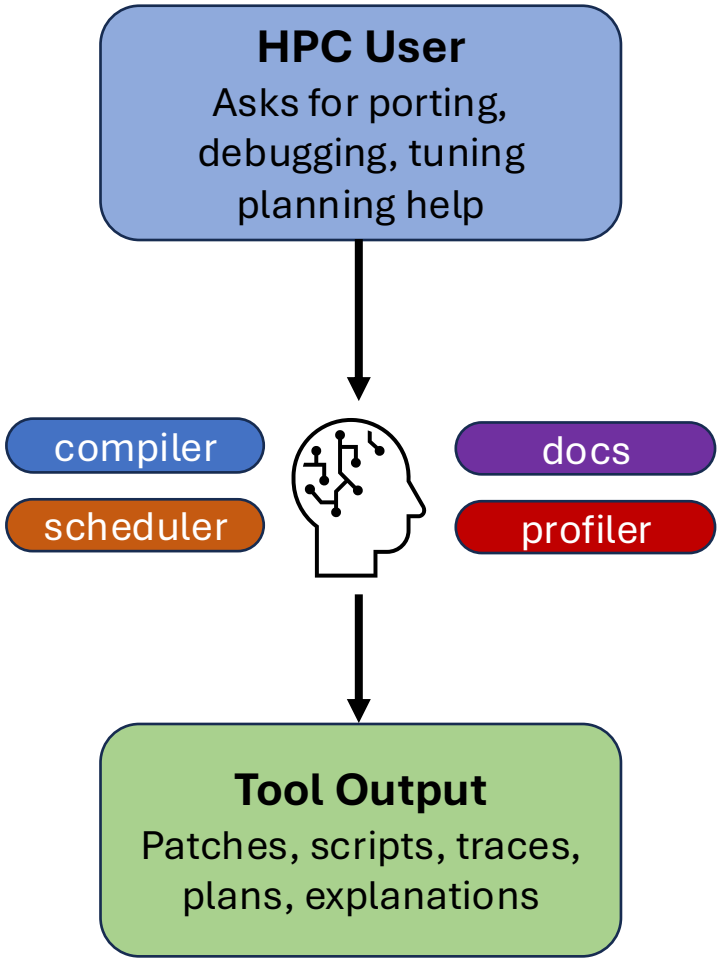
How are LLMs used in experiments: Code Generation

- Generate or rewrite kernels, MPI/OpenMP/CUDA code, job scripts, input decks, compiler flags, launch parameters, or workflow configurations.



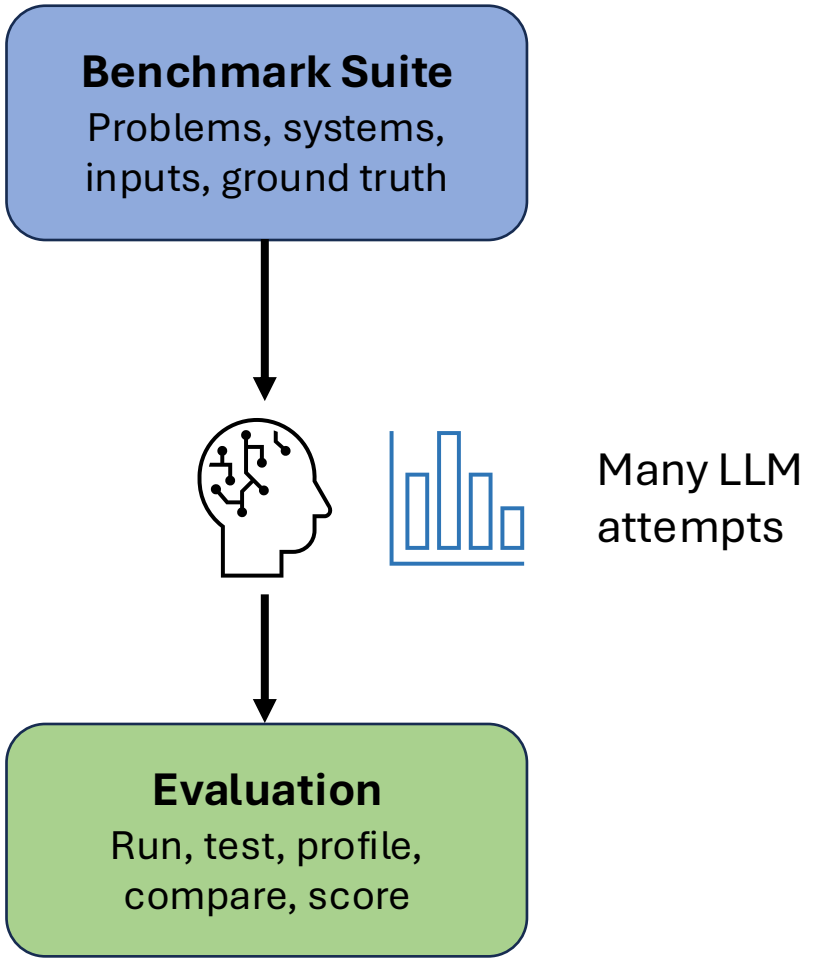
How are LLMs used in experiments: Tools

- LLMs are embedded into HPC tools or agents for tasks such as code porting, performance debugging, job-script generation, experiment planning, or workflow automation.



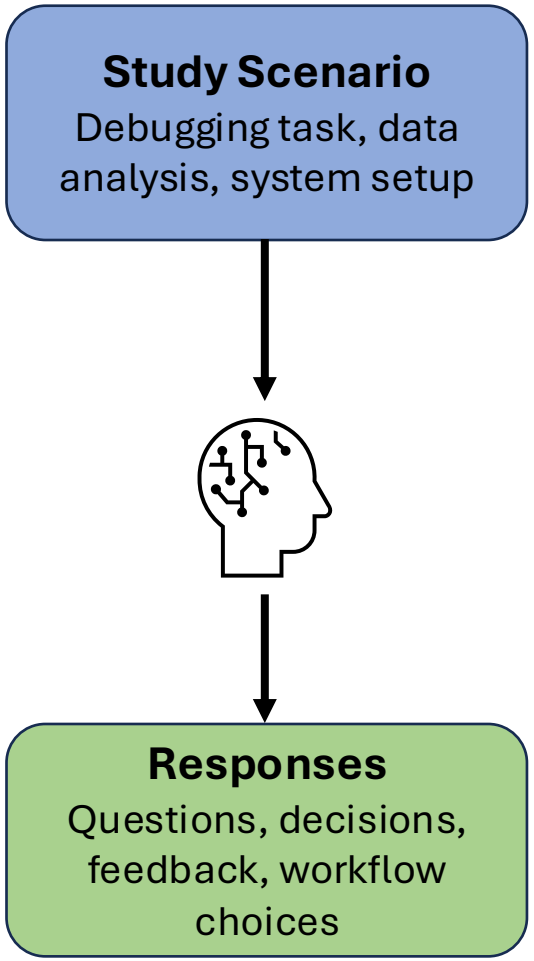
How are LLMs used in experiments: Benchmarking

- Benchmarks measure LLM performance on standardized HPC tasks with reference problems, datasets, systems, and metrics.



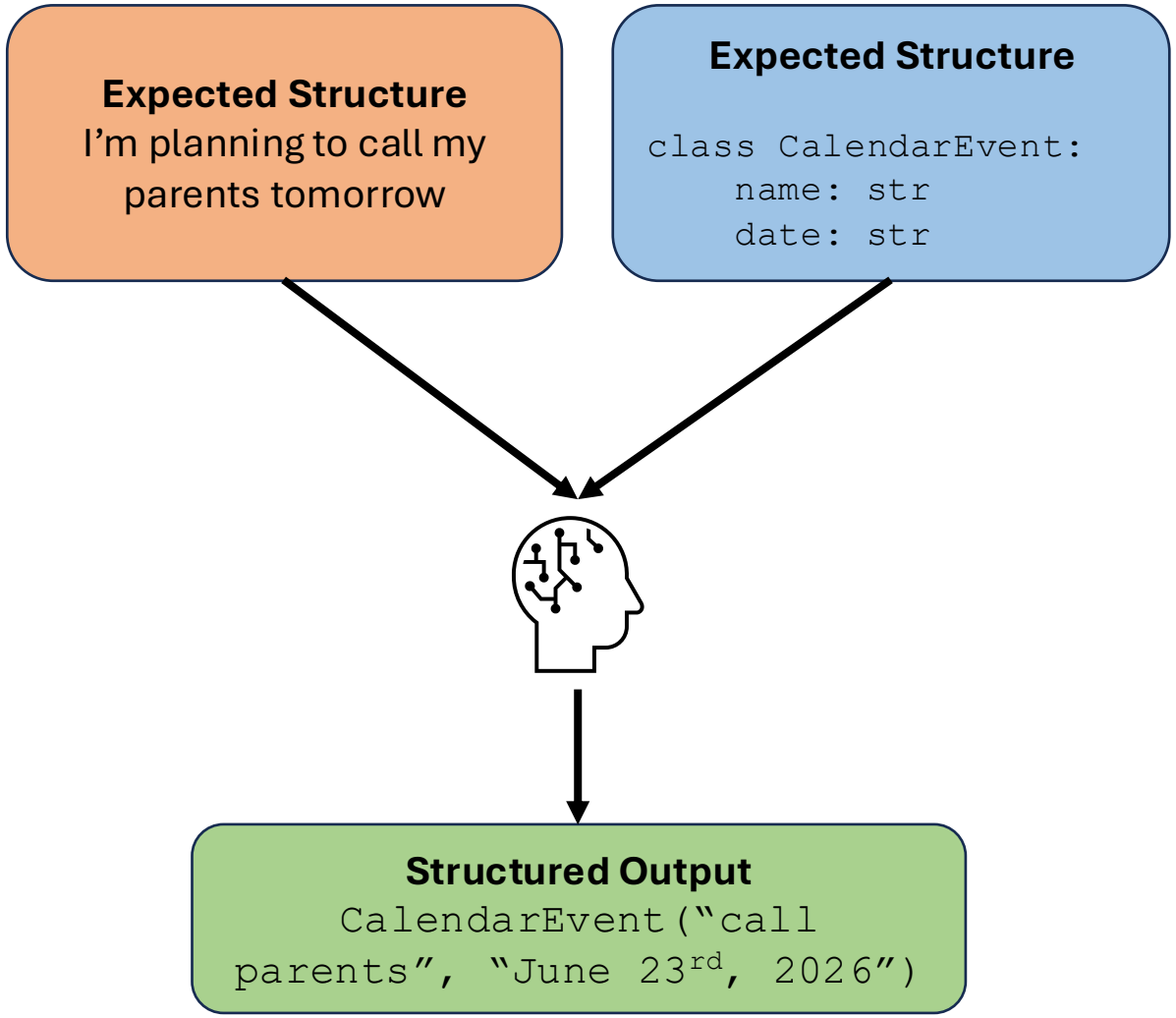
How are LLMs used in experiments: Subjects

- LLMs act as simulated HPC users, domain scientists, performance engineers, or system operators in studies of workflows and interfaces.



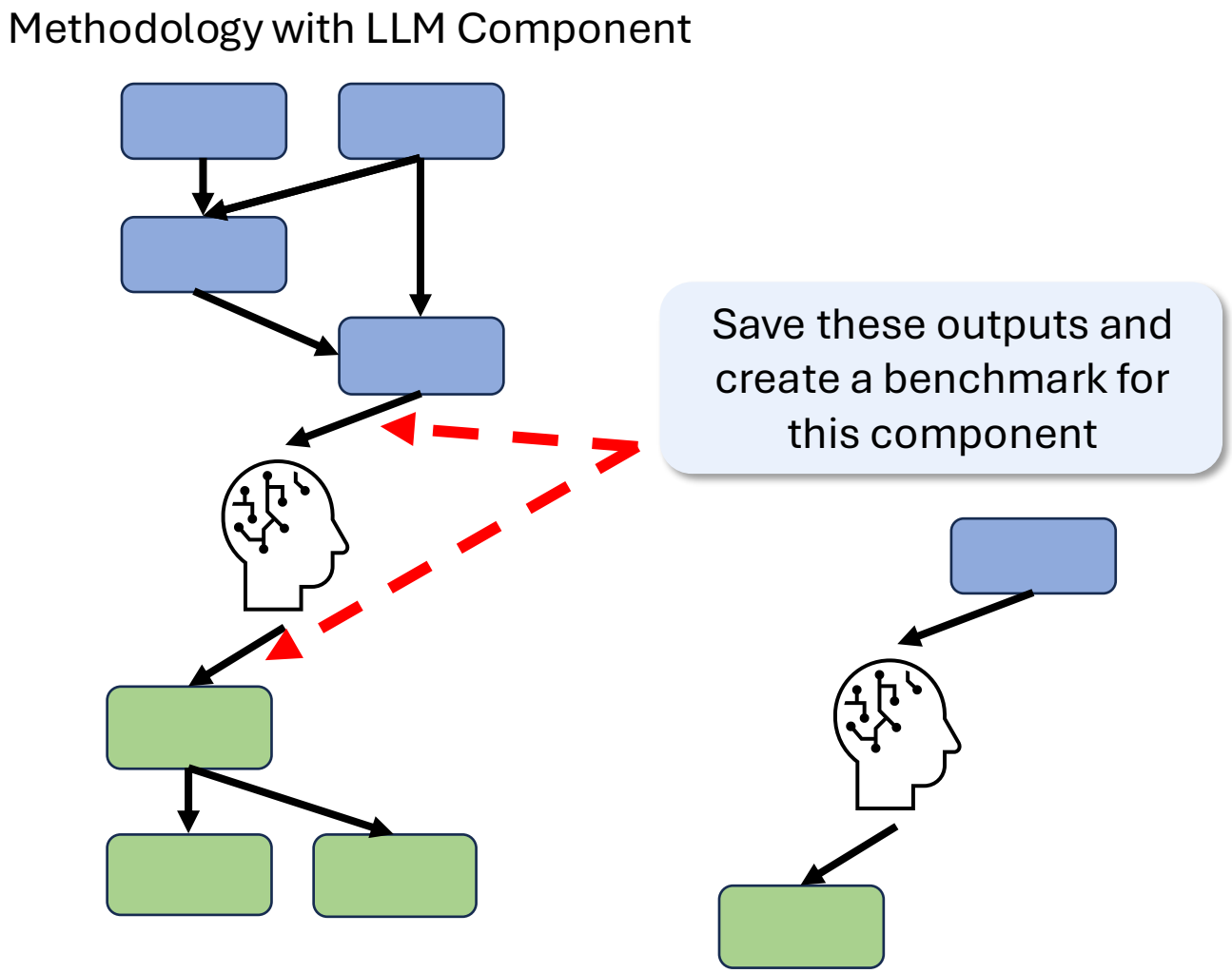
Structured Inputs/Outputs: Mitigating Stochasticity

- Most LLM/Agent frameworks support structured outputs
- Force the LLM to output JSON with particular fields
- Errors or retries if invalid structure
- Reduces potential randomness in outputs



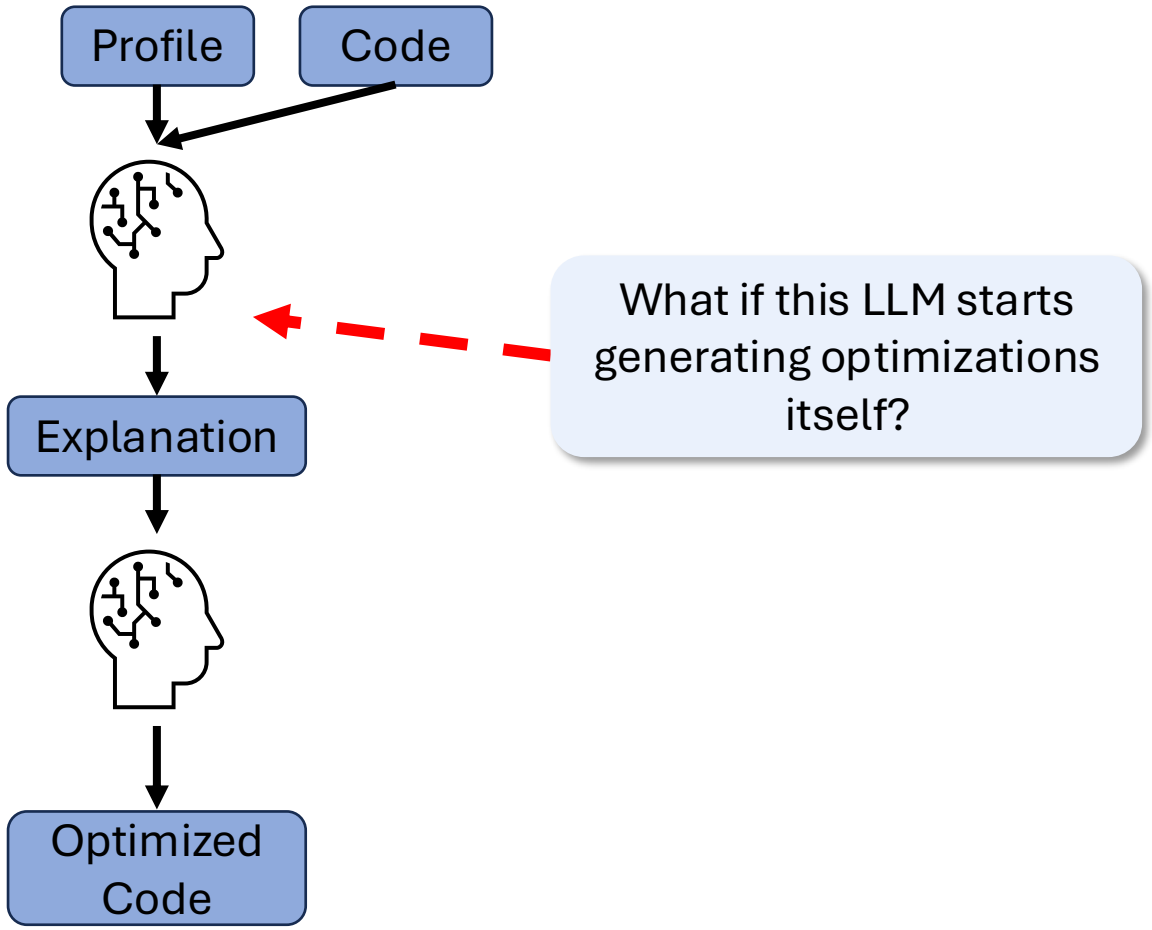
Testing Individual Components and Ablations

- Log inputs and outputs to agents
- Create a benchmark using logged data
- Evaluate standalone components



Human Validation

- Even if human validation is not part of the experimental method, you should sanity check your outputs!



Section V: Transparency and Reproducibility

- Goal for this section: how to ensure LLM subcomponents are reproducible

- Overview
  - Reproducibility requires transparency
  - What to report
  - Logging LLM inputs/outputs
  - Anonymizing LLM artifacts for peer review

Reproducibility and Transparency

- LLMs are stochastic
  - There are ways to make them reproducible, but it is difficult or impossible for large open models or commercial models
  - Thus, generated outputs need to be part of the artifact to make them reproducible
- An independent researcher should be able to reproduce your research by “replaying” generated outputs, instead of generating them again

Logging Generations and Making Them Available for Peer Review

- APIs
  - Simply log API outputs or wrap API calls in a logging function
- Agents
  - Most, if not all, agent frameworks provide tracing functionality
  - Use agent traces to save sequence of actions and generations
- Anonymizing agent traces
  - Anonymizing requirements vary by venue
  - Tools like anonymous4science can host files and mask out provided key words – it provides a link that can be included in a paper or artifact submission

Logging Agent Calls

- OpenAI Agents SDK
  - Traces are logged by default; OPENAI_AGENTS_DISABLE_TRACING=1
- LangChain
  - LANGSMITH_TRACING=true
- LlamaIndex
  - mlflow.llama_index.autolog() use MLFlow to log
- AutoGen
  - Uses Python logging

Demo 2: "Replaying" Logged Generations

```
client = OpenAI(http_client=httpx.Client(
                transport=RecordingTransport(cache_dir="…"))
)

# Transport intercepts every API request
class RecordingTransport(httpx.BaseTransport):
        def handle_request(self, request):
                key = hash_request(request)

                # Replay mode: return saved response
                if REPLAY:
                        return load_response_from_log(key)

                # Normal mode: call OpenAI
                response = real_transport.handle_request(request)
                body = response.read()

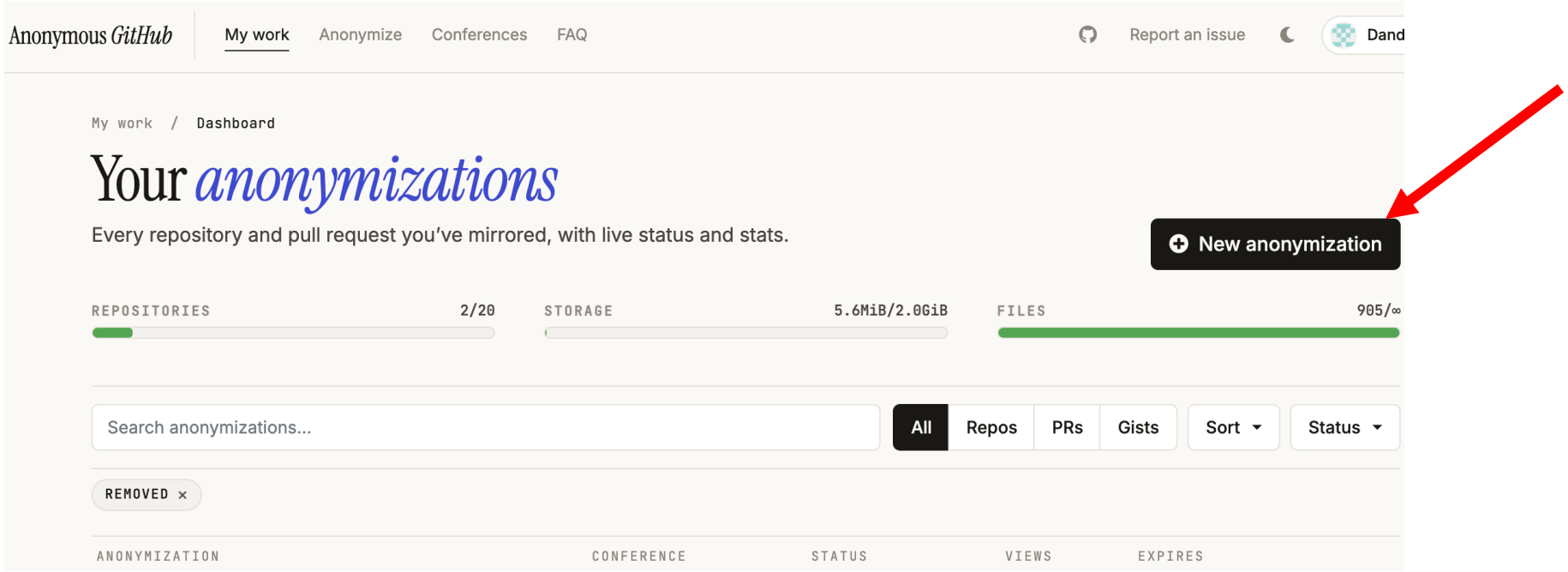
                # Log request + response
                save_json(key, response)
                return rebuild_response(response, body)
```

It only takes tens of lines of code to intercept API calls, log them, and/or replay them

See replaying API calls demo

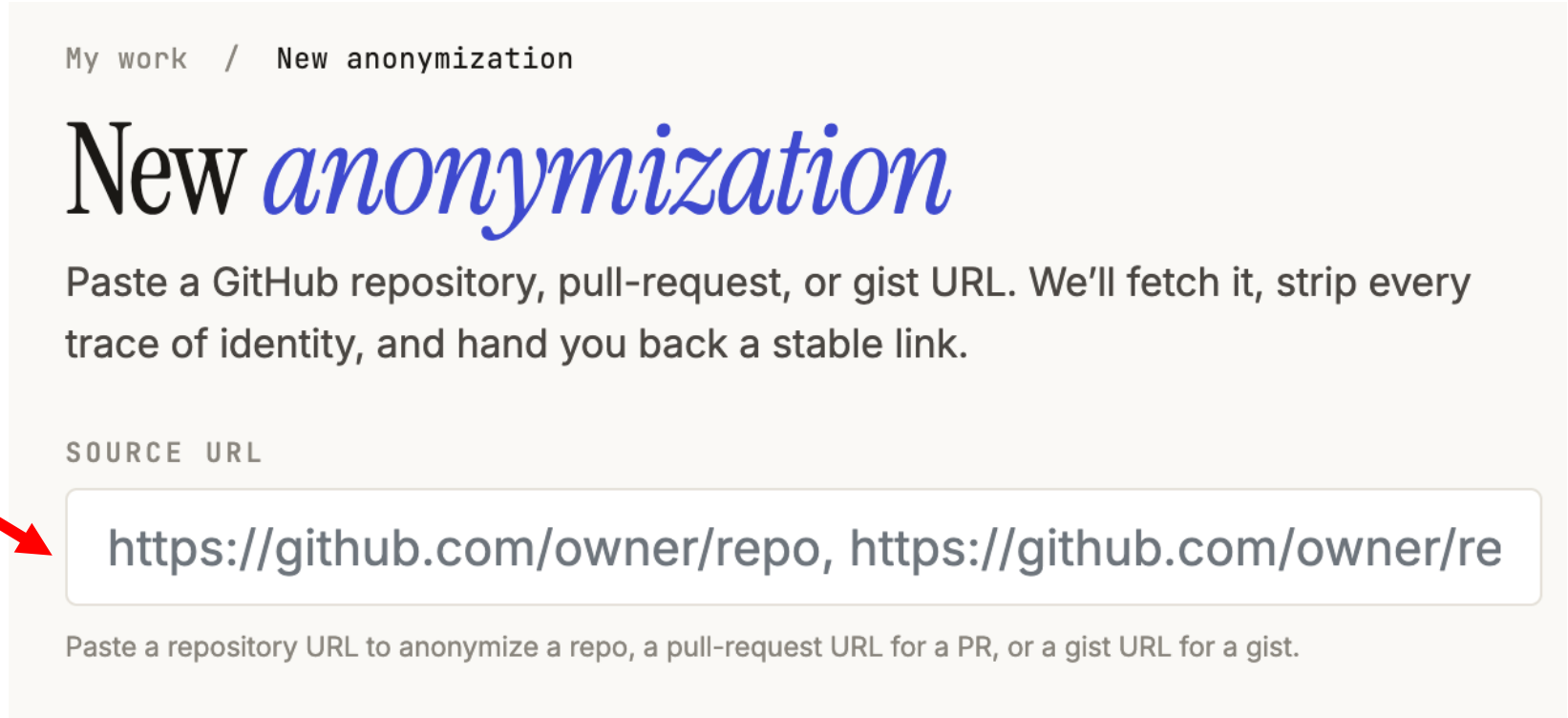
Hosting Anonymous Data with Anonymous4science

- https://anonymous.4open.science/



Hosting Anonymous Data with Anonymous4science

- https://anonymous.4open.science/



Ways to Report

- In Paper/Report/Technical Document
  - Components and decisions crucial to research methodology
- In Supplemental Material / Artifact
  - More space to present prompts and other accompanying material

Types of Reporting

- Declaration
- Models/Versions
- Design
- Traces
- Benchmarks & Metrics
- Reference: Human and OpenLLM
- Limitations

Types of Reporting: Declaration

- The important part of disclosing LLM usage
- All LLM usage in research should be disclosed
- Disclosure requires specifics on where and why an LLM is used
- If the LLM is integral to the research, disclosure should be noticeable

Types of Reporting: Models/Versions

- The model, version, quantization, software versions, etc. must all be reported
- Sampling parameters must be reported as defaults can change over time
- Report why selections were made

Types of Reporting: Design

- What is the workflow and design of the LLM system
- How does data flow from component to component
- Consider Codex vs Claude Code vs ChatGPT: all construct prompts and tools very differently
- Prompts are part of design

Types of Reporting: Traces

- What are the outputs from the LLMs and Agents
- Ideally, representative samples are included in the paper/report
- Traces are essential for reproducibility

Types of Reporting: Benchmarks & Metrics

- Benchmarks and metrics need to be motivated in papers and reports
- All benchmark and experiment configurations, eg number of repetitions, need to be reported
- Report how the LLM was used on the benchmark

Types of Reporting: Reference

- Humans and Open-Source LLMs are important context for how well an LLM approach is doing
- Prefer to include one or the other if using commercial frontier LLMs
- Commercial-only model results are not reproducible for most

Types of Reporting: Limitations

- Like most other HPC studies it is important to discuss limitations
- LLMs present novel, unique challenges to modern HPC research

Reporting Checklist

✓ - must have
OK – should have

	Annotate	Judge	Synthesis	Generate	Tools	Benchmark	Subject
Declare	✓	✓	✓	✓	✓	✓	✓
Model/Versions	✓	✓	✓	✓	✓	✓	✓
Design	✓	✓	✓	✓	✓	✓	✓
Traces	OK	OK	OK	OK	OK	OK	OK
Metrics	✓	OK	OK	OK	OK	✓	OK
Reference	OK	OK	OK	OK	OK	OK	OK
Limitation	✓	✓	✓	✓	✓	✓	✓

Inspired by “LLM Guidelines”, Baltes et al

Section VI: Human Validation: When and How

- Goal for this section: how should humans interact with LLM research components

- Overview
  - When to keep humans in the loop
  - How to employ human evaluation on LLM subcomponents

Issues with Human Validation

- Humans are inconsistent scorers
  - Even the same person may score things differently across time
- It is not reproducible
- Humans are very easily swayed; it is easy to measure the wrong metric
  - Example: people often find longer, more technical answers better regardless of their correctness

An Example: Which is Better?

Can this loop be safely parallelized with OpenMP? Explain briefly.

```
for (int i = 1; i < n; i++) {
    a[i] = a[i - 1] + b[i];
}
```

A: Yes. Each iteration writes to a distinct element a[i], so there is no write-write race. A suitable OpenMP version is:

```
#pragma omp parallel for
for (int i = 1; i < n; i++) {
    a[i] = a[i - 1] + b[i];
}
```

This should improve performance for large n.

B: No, not safely as written. Although each iteration writes a distinct a[i], iteration i reads a[i - 1], which may have been produced by iteration i - 1.
That creates a loop-carried dependence: `a[i - 1] -> a[i]`

A naive #pragma omp parallel for can change the result.

When and How to Use Human Validation

- Use as a supplement to reproducible metrics
- All human scoring and raw results need to be publicly available
  - This is part of the methodology now
- Use metrics that are less susceptible to subjectivity
  - Example: rather than ask “Which output summarizes the scientific result better?”, instead ask “Based on the following generated output, please answer the following multiple-choice questions about a scientific result?”
  - Provide a reproducible metric

Good Human Validation

Explaining Performance Profiles

The code represented in the profile is memory-bound. This is evidenced by the...



Human:

Vote
(A) Good explanation
(B) Ok Explanation
(C) Bad Explanation

Explaining Performance Profiles

The code represented in the profile is memory-bound. This is evidenced by the...



Human:

Please use the explanation to optimize the given code.

> code input

Section VII: Baselines, Metrics, and Ablations

- Goal for this section: best practices for experimental setups for LLM-based research

- Overview
  - What's a good baseline
  - Metrics for LLMs

LLM Baselines

- Representative baselines are important!
- Compare with models of similar: cost, performance (time), quality, availability
- Compare with frontier models / agents
  - Being the best open model is valuable research
  - But comparison to SotA is still necessary context

Probabilistic Metrics

- Since LLMs are random, we care about *how often* they are correct
- For every evaluation using an LLM we need to gather many samples to compute a probabilistic metric
- Common examples:
  - Harmonic and geometric means
  - metric@k e.g. pass@k

@k Metrics For Binary Outputs

- For **binary outputs** @k style metrics measure the probability an LLM gets **at least** one particular output type **given *k* attempts**.
- Example
  - We are evaluating the rate at which LLMs write code with *race conditions*
  - For a given coding task, let *race@k* be the probability that, if the LLM has *k* attempts to write the code, at least one of the *k* will have a race condition
  - Generate $N \gg k$ outputs with the LLM

$$\text{race@k} = 1 - \binom{N-r}{k} \Big/ \binom{N}{k}$$

Probabilistic Metrics

- pass@k – How often are outputs correct?

$$\text{pass@}k = \frac{1}{|P|} \sum_{p \in P} \left[1 - \binom{N - c_p}{k} \Big/ \binom{N}{k} \right]$$

N samples generated per prompt

Set of prompts

Number of correct generations

prompt

```
/* Compute the dot product of x and y using
OpenMP */
int product(int *x, int *y, size_t N) {
```

LLM

What is the probability at least one of k responses is correct?

response 1

```
int sum = 0;
#pragma omp parallel for
for (int i = 0; i < N; i++) {
  sum += x[i] * y[i];
}
return sum;
```

response 2

```
int sum = 0;
for (int i = 0; i < N; i++) {
  sum += x[i] * y[i];
}
return sum;
```

⋮

response k

@k Metrics For Numeric Outputs

- Rather than measuring the probability of at least one correct, we instead measure the **expected best** over *k* attempts
- Example
  - We are evaluating register pressure of LLMs generated GPU kernels
  - For a given coding task, let *reg@k* be, if the LLM has *k* attempts to write the code, the expected minimum number of regs of a uniformly sampled subset sized *k*
  - Generate $N \gg k$ outputs with the LLM

$$reg@\mathrm{k} = \sum_{j=1}^{N} \frac{\binom{j-1}{k-1}}{\binom{N}{k}} \cdot reg_j$$

For reg_j in descending sorted order

Measuring Performance of LLM Generated Code

- speedup@k – What's the expected max speedup?

$$\mathbb{E}\left[\max\left\{\frac{T^*}{T_{s_1}}, \ldots, \frac{T^*}{T_{s_k}}\right\}\right]$$

Baseline runtime

Runtime of LLM generations

prompt

```
/* Compute the dot product of x and y
using OpenMP */
int product(int *x, int *y, size_t N) {
```

LLM

What is the *expected max speedup* over k response?

response 1

```
int sum = 0;
#pragma omp parallel for
for (int i = 0; i < N; i++) {
   sum += x[i] * y[i];
}
return sum;
```

response 2

```
int sum = 0;
for (int i = 0; i < N; i++) {
   sum += x[i] * y[i];
}
return sum;
```

⋮

response k

Measuring Performance of LLM Generated Code

- speedup@k – What's the expected max speedup?

Baseline sequential runtime

$$\mathbb{E}\left[\max\left\{\frac{T^*}{T_{s_1,n}},\ldots,\frac{T^*}{T_{s_k,n}}\right\}\right]$$

Runtime of LLM generations on *n* processors

prompt

```
/* Compute the dot product of x and y
using OpenMP */
int product(int *x, int *y, size_t N) {
```

LLM

response 1

```
int sum = 0;
#pragma omp parallel for
for (int i = 0; i < N; i++) {
   sum += x[i] * y[i];
}
return sum;
```

response 2

```
int sum = 0;
for (int i = 0; i < N; i++) {
   sum += x[i] * y[i];
}
return sum;
```

⋮

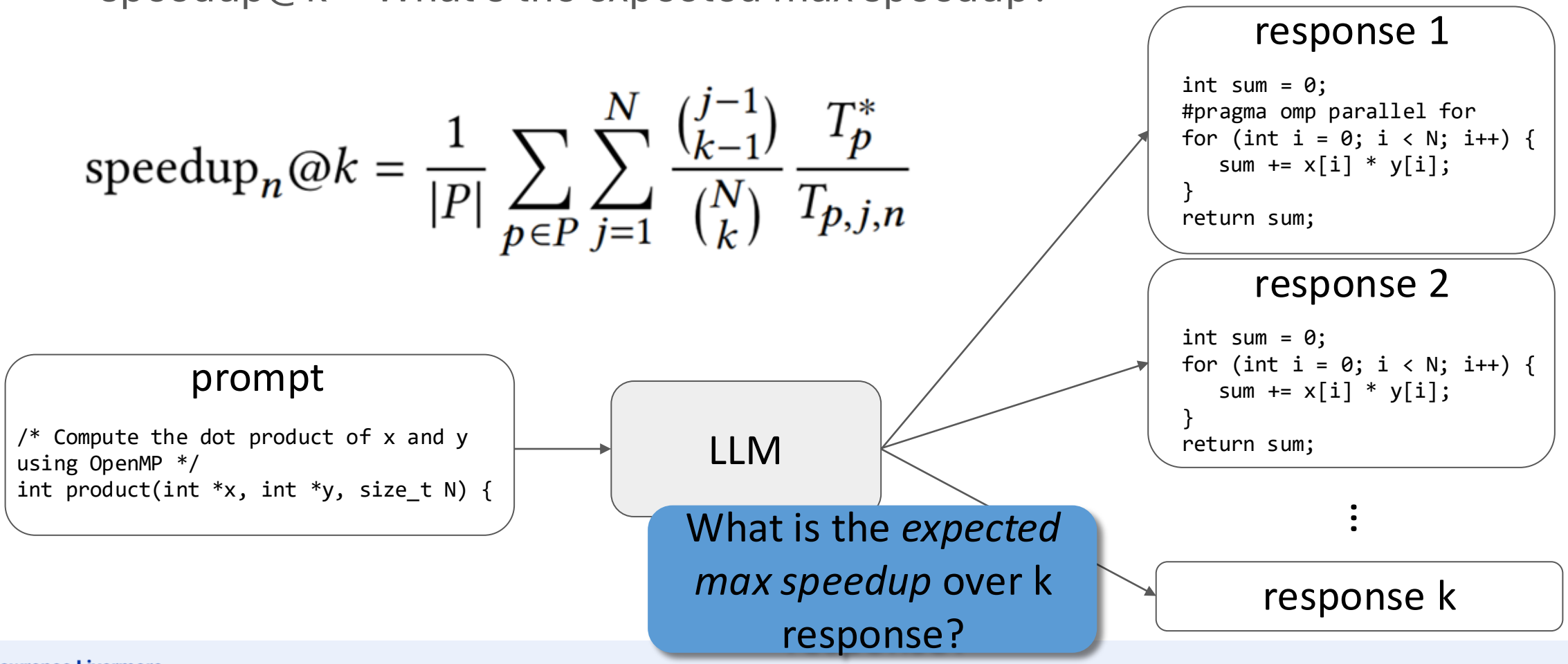
response k

What is the *expected max speedup* over k response?

Measuring Performance of LLM Generated Code

- speedup@k – What's the expected max speedup?

$$\text{speedup}_n@k = \frac{1}{|P|} \sum_{p \in P} \sum_{j=1}^{N} \frac{\binom{j-1}{k-1}}{\binom{N}{k}} \frac{T_p^*}{T_{p,j,n}}$$



LLMs and HPC Research

- Where do you see problems arise using LLMs in HPC research?
- What tooling/software would make it easier for you to develop more reproducible agents?

Conclusion

- Questions?